



Database Programming with PL/SQL

1-1

Introduction to PL/SQL



Objectives

This lesson covers the following objectives:

- Describe PL/SQL
- Differentiate between SQL and PL/SQL
- Explain the need for PL/SQL

Purpose

- PL/SQL is Oracle Corporation's procedural programming language for relational databases.
- To describe PL/SQL, you learn its characteristics and identify the differences between PL/SQL and SQL.
- Identifying limitations of SQL and how PL/SQL addresses those limitations will help you to understand why PL/SQL is needed.

PL/SQL Description

- Procedural Language extension to SQL.
- A third-generation programming language (3GL).
- An Oracle proprietary programming language.
- Combines program logic and control flow with SQL.



Evolution/Generations of Programming Languages:

- 1GL: First-generation programming languages; these are machine level languages specific to a particular CPU
- 2GL: Second-generation programming languages; assembly languages specific to a particular CPU; converted by an assembler into a machine language; commonly used for performance-oriented and processing-intensive applications such as firmware interfaces and hardware drivers
- 3GL: Third-generation programming languages; converted into machine language by a compiler; less cryptic and thus more programmer-friendly than 2GLs (ex., Visual Basic, C, C++, COBOL, FORTRAN, Java, Pascal, PL/SQL)

Evolution/Generations of Programming Languages:

- 4GL: Fourth-generation programming languages; less cryptic and thus more programmer-friendly than 2GLs; unlike the broad applicability of 3GLs, most 4GLs are used with databases, for queries, report generation, data manipulation, etc. (ex., SQL, MySQL)
- 5GL: Fifth-generation programming languages; generally involve a visual or graphical development environment that exports source language to a 3GL or 4GL compiler; 5GL may also refer to languages that define a problem and a set of constraints, then let the computer find a solution; (ex., AI systems such as PROLOG used with IBM's Watson)

As an Oracle proprietary programming language, PL/SQL is only used with an Oracle database.



SQL Description

- A nonprocedural language.
 - Also known as a "declarative language," allows programmer to focus on input and output rather than the program steps.
- A fourth-generation programming language (4GL).
- Primary language used to access and modify data in relational databases.
- Standardized by the American National Standards Institute (ANSI).
- Vendors such as Oracle typically include some proprietary SQL features in their database environments.

SQL Statement

- The SQL statement shown is simple and straightforward.
- However, if you need to modify a data item in a conditional manner, you come across a limitation of SQL.

```
SELECT employee_id, job_id, hire_date  
FROM employees;
```

- For example, how would you write an SQL statement to update the job_id data with a new value determined by the current job_id and the hire_date?

Limitations of SQL

- Assume the company decides to promote all sales representatives, marketing representatives, and stock clerks employed for at least ten years to senior representatives and clerks.
- If the current date is 05-Feb-2015, sales representatives 174, 176, and 178 qualify for the promotion.

EMPLOYEE_ID	JOB_ID	HIRE_DATE	“NEW” JOB_ID
174	SA_REP	11-May-1996	SR_SA_REP
176	SA_REP	24-Mar-1998	SR_SA_REP
178	SA_REP	24-May-1999	SR_SA_REP
240	SA_REP	02-Oct-2005	
242	SA_REP	09-Dec-2007	

Limitations of SQL

If the current date is 05-FEB-2015, stock clerks 141, 142, 143, and 144 also qualify for the promotion.

EMPLOYEE_ID	JOB_ID	HIRE_DATE	"NEW" JOB_ID
141	ST_CLERK	17-Oct-1995	SR_ST_CLERK
142	ST_CLERK	29-Jan-1997	SR_ST_CLERK
143	ST_CLERK	15-Mar-1998	SR_ST_CLERK
144	ST_CLERK	09-Jul-1998	SR_ST_CLERK
244	ST_CLERK	07-Sep-2009	

Limitations of SQL

- One solution to updating the job_id data is shown.
- How many SQL statements do you need to write for sales representatives, marketing representatives, and stock clerks?
- What if there are other job_ids to update?

```
UPDATE employees
  SET job_id = 'SR_SA_REP'
 WHERE job_id = 'SA_REP' AND
        hire_date <= '05-Feb-2005'
```

```
UPDATE employees
  SET job_id = 'SR_ST_CLERK'
 WHERE job_id = 'ST_CLERK' AND
        hire_date <= '05-Feb-2005'
```

and so on...

Limitations of SQL

- You would need to write a separate SQL statement for each `job_id` that needs to be updated.
- Depending on the number of `job_ids`, this could be a tedious task.
- It would be easier to write a single statement to accomplish this task.
- The statement would require logic, otherwise known as procedural logic.
- PL/SQL extends SQL with procedural logic and makes it possible to write one statement to accomplish this task.

PL/SQL Extends SQL with Procedural Logic

Using PL/SQL, you can write one statement to promote the sales representatives, marketing representatives, and stock clerks.

```
DECLARE
  CURSOR c_employees IS SELECT * FROM employees;
BEGIN
  FOR c_emp IN c_employees
  LOOP
    IF c_emp.job_id = 'SA_REP' AND c_emp.hire_date <= '05-Feb-2005' THEN
      UPDATE employees SET job_id = 'SR_SA_REP' WHERE employee_id =
c_emp.employee_id;
    ELSIF c_emp.job_id = 'MK_REP' AND c_emp.hire_date <= '05-Feb-2005' THEN
      UPDATE employees SET job_id = 'SR_MK_REP' WHERE employee_id =
c_emp.employee_id;
    ELSIF c_emp.job_id = 'ST_CLERK' AND c_emp.hire_date <= '05-Feb-2005' THEN
      UPDATE employees SET job_id = 'SR_ST_CLRK' WHERE employee_id =
c_emp.employee_id;
    END IF;
  END LOOP;
END;
```

Procedural Constructs

- You use PL/SQL to write the procedural code and embed SQL statements within the PL/SQL code.
- The procedural code includes variables, constants, cursors, conditional logic, and iteration.
- PL/SQL code blocks can be saved and named, then executed whenever needed.



Procedural Constructs Highlighted

Several PL/SQL constructs are highlighted below.

```
DECLARE
  Cursor
  CURSOR c_employees IS SELECT * FROM employees;
BEGIN
  FOR c_emp in c_employees
  LOOP
    IF c_emp.job_id = 'SA_REP' AND c_emp.hire_date <='05-Feb-2005' THEN
      UPDATE employees SET job_id = 'SR_SA_REP'
        WHERE employee_id = c_emp.employee_id;
    ELSIF c_emp.job_id = 'MK_REP' AND c_emp.hire_date <= '05-Feb-2005'
      THEN UPDATE employees SET job_id = 'SR_MK_REP'
        WHERE employee_id = c_emp.employee_id;
    ELSIF c_emp.job_id = 'ST_CLERK' AND c_emp.hire_date <='05-Feb-2005'
      THEN UPDATE employees SET job_id = 'SR_ST_CLRK'
        WHERE employee_id = c_emp.employee_id;
    END IF;
  END LOOP;
END;
```

Iterative Control

SQL

Conditional Control

Characteristics of PL/SQL

PL/SQL:

- Is a highly structured, readable, and accessible language.
- Is a standard and portable language for Oracle development.
- Is an embedded language and it works with SQL.
- Is a high-performance, highly integrated database language.
- Is based on the Ada Programming Language and has many similarities in syntax.



Terminology

Key terms used in this lesson included:

- PL/SQL
- Procedural Constructs

Summary

In this lesson, you should have learned how to:

- Describe PL/SQL
- Differentiate between SQL and PL/SQL
- Explain the need for PL/SQL

